

Mechanistic analysis of esophageal corkscrewing driven by active contraction

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Corkscrew esophagus is a pathological condition in which the esophageal wall undergoes helical twisting or localized constrictions in response to strong muscular contractions. To unravel the mechanical origins of this phenomenon, we build a finite-deformation minimal model of the esophageal wall that incorporates an active thin-shell outer layer to capture the effect of contracting muscle fibers. We apply this model to quantify muscle contractility in vivo using high-resolution manometry, and the corkscrew is predicted to emerge as an elastic instability when critical values of contractility are achieved. Analysis of the model and clinical data reveal plausible physiological changes in the esophageal muscle and matrix that precipitate instability including longitudinal muscle impairment and material softening due to repeated straining. Our framework links clinical observations to first-principles mechanics and provides a physically grounded, quantitative criterion for the onset of corkscrew esophagus.

I. INTRODUCTION

Esophageal motility disorders encompass a broad class of conditions that involve difficulty or loss of swallowing, leading to gastroenterological complications. Their pathophysiology is poorly understood but is thought to result from a combination of genetic, neurological, and mechanical factors [1]. Physiological symptoms of esophageal motility disorders vary widely but generally include chest pain, dysphagia, and heartburn [2]. For a particular subset of patients, however, the symptoms can be even more severe: the shape of the esophagus itself changes dramatically. While the esophagus generally appears as a straight or gently curved slender tube, in cases where the contraction is strong, the esophagus has been observed to deform into persistent bent and corkscrewed shapes, as illustrated in Fig. 1A. Because quantifying muscle contractility is challenging to do in vivo, the mechanism that causes the appearance of such striking shapes has remained largely unclear. As a consequence, formulating diagnosis standards can be an extremely daunting task [3, 4], leading to miscommunications, delays, or unnecessary interventions [5–7]. These limitations highlight the need for a more rigorous platform that synthesizes clinical data analysis, first-

principles mechanics, and bioinstrumentation to deliver more precise results.

In this article, we provide a quantitative and physical understanding of the origin of such corkscrew shapes. From a mechanical perspective, this anomalous behavior is reminiscent of classical elastic instabilities [8–16] in which the shape of a solid body changes dramatically after a critical load is attained. Such instabilities play crucial roles in morphogenesis across scales [17–24]. To study the transitions of the esophagus into buckled and corkscrewed shapes, we derive a continuum model for the esophagus that accounts for active stresses due to muscle contraction and show that these phenomena are indeed manifestations of elastic instability when contractility exceeds a certain threshold value. We also describe a way to measure these contraction parameters in situ by combining high-resolution manometry (HRM) [25], electrical impedance [26], and timed barium esophagography (TBE) [27]. Comparison with clinical data reveals parameters that can be used to predict corkscrew esophagus in patients while suggesting physiological changes that make the organ more susceptible to instability. Our results elucidate the role of muscle incoordination in corkscrew formation and disease progression of achalasia.

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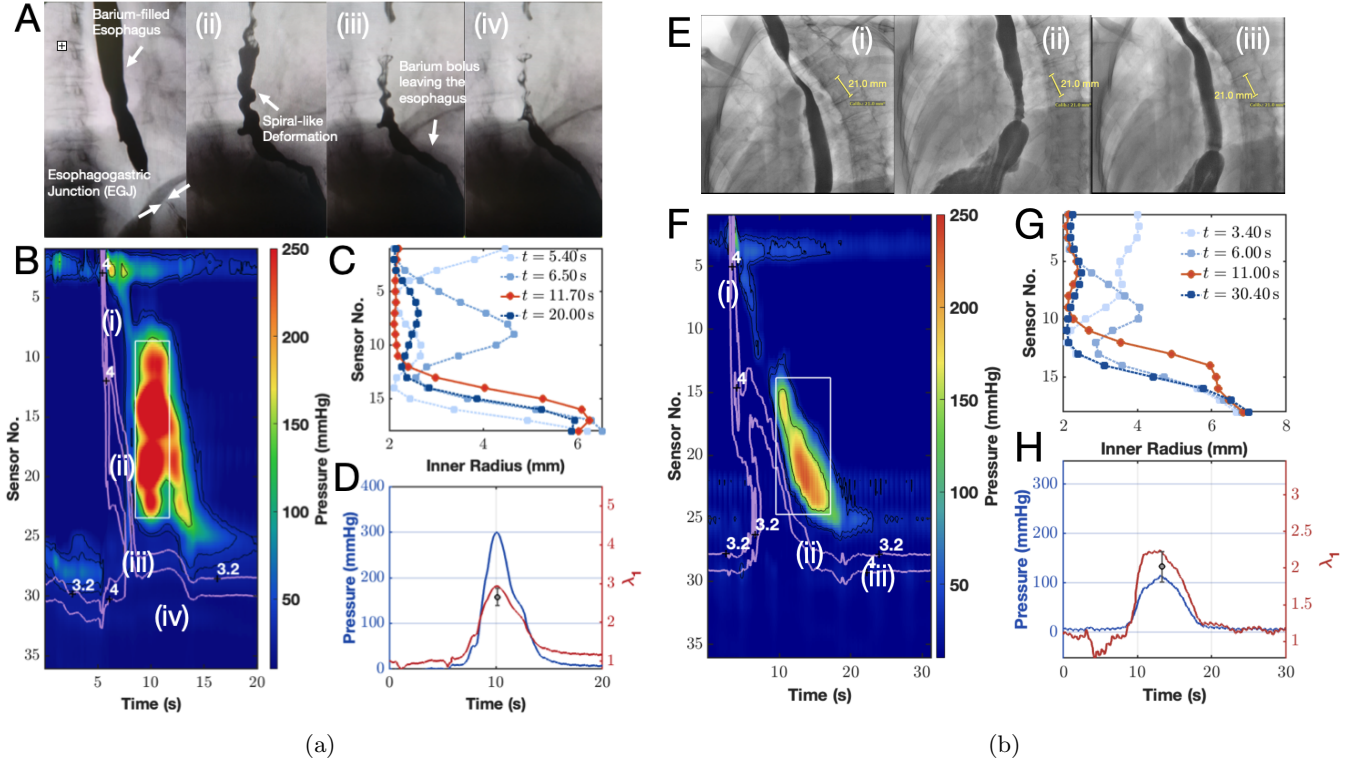


FIG. 1: Left and right columns show representative data from swallows in a corkscrew esophagus patient and a normal subject, respectively. (A) X-ray esophagram images and (B) high-resolution manometry (HRM) recordings are shown for each case; the fluoroscopic images were not acquired simultaneously with the manometry data. Lowercase Roman numerals on the esophagram indicate schematic correspondence to representative phases of the swallow inferred from the manometric sequence. The purple contours denote the luminal radius (in mm) reconstructed from impedance data [26]. (C) Lumen radius profiles at selected times, showing distension due to a bolus and subsequent relaxation/retention. (D) Spatially averaged luminal pressure and circular muscle contractility parameter λ_1 , as estimated from clinical data. Panels E through H correspond to panels A through D but are for a patient with a normal esophagus.

II. RESULTS

A. Anatomy and Shape of the Esophagus

The esophagus is a hollow, muscular tube, approximately 2 cm in diameter and 25 cm in length in adult humans, that transports food and liquid from the throat to the stomach [28]. As shown in Fig. 2a, the esophagus possesses a layered structure: from the lumen outwards, the walls of the esophagus consist primarily of the mucosa, the submucosa, and two layers of muscular fibers. In the inner layer of muscle, the fibers tend to be oriented circumferentially, whereas in the outer muscle layer, they tend to be oriented longitudinally. During normal peristalsis, these muscle fibers contract in an anisotropic but coordinated way to aid the transport of a bolus down to the stomach. Notably, the relative thickness of the muscular layers tends to be small compared with the thickness of the mucosa and submucosa [29, 30]. The esophagus is approximately a cylinder in its default state (Fig. 2b) and exhibits inherent axial and radial flexibility, significantly changing in length and radius in response to

internal pressures or muscular contractions. However, because of the esophagus's anatomical connections with the pharynx and stomach, the endpoints of the esophagus move only modestly during normal swallowing, and the associated axial shortening is small. For example, one study has reported approximately 7% esophageal shortening during normal peristalsis in rats [31], and, as shown below, the achalasia patients in our cohort without a helical deformation exhibit axial stretch ratios in the narrow range 0.97–1.10.

In this study, the shape of the esophagus was probed using pressure and electrical impedance data from HRM and radiographs from TBE. In HRM, which is often considered the “gold standard” for diagnosis of esophageal disorders [1, 32], a pressure-sensitive catheter is inserted into a patient's esophagus and used to record stresses while swallowing. Each catheter has 36 sensors distributed evenly along its length, and each sensor can detect the stress acting on it, be it due to gas inside the lumen, a fluid bolus, or direct contact with the esophageal wall. The pressures over time and the length of the esophagus are recorded as colored kymographs, such as the ones shown in Fig. 1B. Certain patterns in manometric

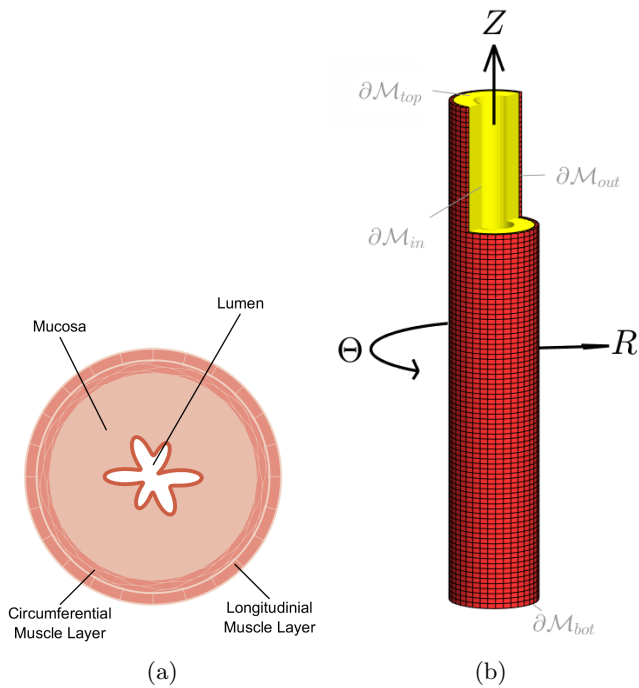


FIG. 2: (Left) Schematic of a cross-section of the esophagus.

From inside to outside: the lumen, which contains a pressurized gas or a bolus; the mucosa, a thick but soft layer of tissue; the circumferential muscle (CM) layer; and the longitudinal muscle (LM) layer. (Right) Schematic of the model geometry, consisting of a cylinder with four boundary components $\partial\mathcal{M}_{in}$, $\partial\mathcal{M}_{out}$, $\partial\mathcal{M}_{top}$, and $\partial\mathcal{M}_{bot}$, representing the esophagus in its reference configuration. The passive mucosal/submucosal layer (yellow) is enclosed by an actively contracting muscular layer (red). In the muscular layer, the fibers (black lines) run both circumferentially and longitudinally. Part of the cylinder has been cut away to show the cross-section and inner and outer boundaries.

data can be indicative of disease; for example, a wide region of warmer colors near the bottom of the plot can signify an elevated pressure at the LES, which tends to be indicative of achalasia or esophagogastric junction (EGJ) outlet obstruction [1]. A given patient typically has data from a dozen or more swallows taken under a variety of conditions.

It is also possible to take electrical impedance data concurrently with pressure data during manometry trials. Impedance data can be converted into a measurement of lumen radius using an established relationship between bolus conductivity and lumen cross-sectional area [26]. Each manometer is equipped with 18 electrodes, which detect the electrical impedance of the esophagus (Fig. 1B, purple contours). The degree of dilatation or contraction of the lumen varies throughout the swallowing process (Fig. 1C); notably, during peak contraction, the esophageal wall is in contact with the manometer (radius = 2.1 mm).

As a complement to manometry, TBE studies were performed to provide a direct visualization of the shape

of the esophagus (Fig. 1A). This technique is useful for identifying abnormal esophageal contraction or emptying and is particularly helpful for confirming the presence of corkscrew esophagus, as in Fig. 1A. It also allows one to observe more directly the physical dimensions of the esophagus, though care must be taken since the esophagus deforms as a result of the fluid pressure. We will show that the data taken from HRM and TBE can be used to estimate muscle contractility within the esophagus (Fig. 1D). For normal patients, TBE, HRM, impedance, and contractility data tend to be qualitatively very different than for corkscrew and achalasia patients (Fig. 1E–H).

B. Presentation of Achalasia and Corkscrew Esophagus

Achalasia is an esophageal motility disorder characterized by impaired relaxation of the lower esophageal sphincter, which inhibits the egress of food and liquid [33]. A signature of achalasia is the lack or failure of peristalsis. According to the widely used Chicago Classification system [1, 34], there are three subtypes of achalasia with markedly different clinical characteristics; however, the presentations of different types of achalasia are often similar, leading to challenges in diagnosis and treatment. The type of achalasia is loosely determined by the level of esophageal pressurization, with type III achalasia associated with high lumen pressures and strong, spastic contractions during swallowing, type I achalasia associated with lack or near lack of pressurization and contractility altogether, and type II achalasia associated with uniform “panesophageal” pressurization of an intermediate strength. It has been hypothesized that the different achalasia types represent stages of disease progression in which the muscles of the esophagus are initially hyperactive but gradually decompensate, ultimately yielding a compliant and dilated esophagus [35, 36].

A subset of achalasia III patients exhibit what is known as corkscrew esophagus, in which the esophagus loses its cylindrical shape and becomes long and tortuous. Physiologically, patients diagnosed with corkscrew esophagus frequently present with significant axial elongation of the esophagus compared with those of healthy controls, as can be seen in Fig. 1, or compared with achalasia III patients who do not have corkscrew esophagus, as shown in Fig. 5. The radius of the lumen is also significantly larger for corkscrew patients compared with noncorkscrew achalasia III patients, as shown in Fig. 6. Throughout this work, we classify a patient as having a corkscrew esophagus if a nonzero number of helical turns is observed on the esophagram. The number of turns varies from 1 to 7 among our patients and is most commonly 5. There is no obvious preference in the handedness of the spiral. Videos taken during TBE confirm that corkscrew esophagus is a transient phenomenon that occurs while the muscles of the esophagus are actively con-

tracting; upon relaxation, the corkscrew shape is significantly diminished or disappears. Corkscrew esophagus is associated with achalasia III, but persistent corkscrew-like structures are also observed in patients with other esophageal diseases such as diffuse esophageal spasm [37] or nutcracker esophagus [38] in which muscle contractility can also be abnormally high.

Clinical observations indicate that the corkscrew pattern is frequently accompanied by an obstruction at the EGJ. This is supported by the following evidence. First, manometric data (Fig. 1) reveal that the corkscrew is almost always accompanied by unusually strong, simultaneous pressure spasms, often referred to as panesophageal contractions [33]. Additionally, for patients with type III achalasia, a distal myotomy alone can resolve the corkscrew pattern, even if the corkscrew morphology itself is proximal. Finally, historical cohorts treated with pneumatic dilation also showed substantial improvement in corkscrewing, supporting the idea that reducing EGJ resistance can eliminate the spiral morphology even without changing the esophageal body musculature. Therefore, EGJ obstruction may lead to a bolus being trapped, which in turn causes a persistent elevated intraluminal stress as the esophageal muscle contracts in an attempt to expel the bolus, allowing tertiary or distorted contractions to become morphologically spiral. Motivated by these observations, we treat enhanced contractile strength as a downstream consequence of EGJ obstruction rather than modeling the obstruction explicitly.

C. Two-Component Finite-Deformation Continuum Model

We employ continuum mechanics to predict the shape of the esophagus when subjected to various stresses. Hyperelastic continuum models remain popular for modeling nonlinearly compliant gastrointestinal tissues [39] and can readily be modified to include active stresses or strains [30, 40–43]. We elect to work in the reference (i.e., Lagrangian) configuration owing to its simple geometry, which can be easily described using conventional cylindrical coordinates (R, Θ, Z) , illustrated in Fig. 2b, with corresponding orthonormal frame $\{\mathbf{E}_R, \mathbf{E}_\Theta, \mathbf{E}_Z\}$. Our reference configuration consists of a hyperelastic three-dimensional cylinder $\mathcal{M}$ of radii R_i and R_o (where $0 < R_i < R_o$), representing the combined mucosa and submucosa, as well as a shell of thickness $\delta R_o \ll R_o$ wrapped around the outer boundary of $\mathcal{M}$ where $R = R_o$, representing the muscular layers. In positing this model, we have made use of the fact that the thickness of the muscular layer is generally much smaller than that of the mucosa and submucosa [44]. To maintain simplicity and generality, we model the mucosa as an incompressible neo-Hookean solid with elastic modulus μ . The muscle layer is similarly taken to be neo-Hookean and incompressible and has its own (three-dimensional) modulus $\bar{\mu}$. In our analysis, we employ a quasistatic approximation

since the persistence time of the corkscrew is significantly longer than the time scale of swallowing and other deformations. Under such conditions, the body $\mathcal{M}$ will deform in such a manner that minimizes the integral of the total strain energy function $\int dV (W_e + \bar{W}_e)$, where

$$W_e = \frac{\mu}{2} \text{tr}(\mathbf{F}^T \mathbf{F}) - p(\det \mathbf{F} - 1) \quad (1)$$

$$\bar{W}_e = \frac{\bar{\mu}}{2} \text{tr}(\bar{\mathbf{F}}^T \bar{\mathbf{F}}) - \bar{p}(\det \bar{\mathbf{F}} - 1). \quad (2)$$

Here, $\mathbf{F}$ and $\bar{\mathbf{F}}$ are the deformation gradients of the mucosa and muscle layers, respectively, and p and $\bar{p}$ are the hydrostatic pressures in the mucosa and muscle, respectively, enforcing the constraints of incompressibility. The superscript T denotes the matrix transpose. Note that since the layer thickness is taken to be small, the normal stress across the layer in the normal direction is approximately zero [45], which can be used to determine $\bar{p}$.

In our model, muscle contractility serves as the primary control parameter governing the emergence of the corkscrew shape. To account for the forces that arise when the circumferential muscle (CM) and longitudinal muscle (LM) fibers contract, we suppose that the elastic sheet is capable of providing an anisotropic active stress $\bar{\mathbf{P}}_a$. Here, we follow [30] and assume that the active stress arises from minimizing the strain energy function

$$\bar{W}_a = \frac{\bar{\mu}_1}{2} \left(\lambda_1 \sqrt{I_{4,1}} - 1 \right)^2 + \frac{\bar{\mu}_2}{2} \left(\lambda_2 \sqrt{I_{4,2}} - 1 \right)^2, \quad (3)$$

where $\bar{\mu}_1$ is the elastic modulus of the circumferential muscular fibers, $\sqrt{I_{4,1}} = [(\mathbf{F}^T \mathbf{F}) : (\mathbf{M}_1 \otimes \mathbf{M}_1)]^{1/2}$ is the new length of a unit tangent vector with orientation $\mathbf{M}_1$ in the reference configuration when deforming to the spatial configuration [46], and λ_1 is a dimensionless parameter that represents the level of contractility in the circumferential fibers. The quantities $\bar{\mu}_2$, $\mathbf{M}_2$, $I_{4,2}$, and λ_2 are defined analogously but for the longitudinal fibers; unless explicitly stated otherwise, $\mathbf{M}_1 = \mathbf{E}_\Theta$ and $\mathbf{M}_2 = \mathbf{E}_Z$. It is possible to identify λ_1^{-1} and λ_2^{-1} with preferred strain values of the fibers. Consistent with clinical observations of panesophageal contractions, λ_1 and λ_2 are taken to be constants in our analysis, though in principle they may depend on time or space [47]. Generally, we assume that λ_1 and λ_2 are greater than or equal to 1 so that the muscle fibers are contracting relative to their base lengths, but in principle this need not be the case.

Define $\mathcal{P}_e = \partial W_e / \partial \mathbf{F}$, $\bar{\mathcal{P}}_e = \partial \bar{W}_e / \partial \bar{\mathbf{F}}$, and $\bar{\mathcal{P}}_a = \partial \bar{W}_a / \partial \bar{\mathbf{F}}$ to be the first Piola–Kirchhoff stress tensors corresponding to W_e , $\bar{W}_e$, and $\bar{W}_a$, respectively. For notational convenience, we define $\mathcal{P} = \mathcal{P}_e$ to be the stress in the bulk and $\bar{\mathcal{P}} = \bar{\mathcal{P}}_e + \bar{\mathcal{P}}_a$ to be the total stress in the layer. Standard variational methods [48] yield the equations of equilibrium for our system (a complete derivation can be found in the Supporting Information), which are

$$\text{Div } \mathcal{P} = 0 \quad (4)$$

in the mucosa (note that we have assumed that body forces vanish) and the boundary condition

$$\delta R_o \overline{\text{Div}} \bar{\mathbf{P}} = \mathbf{P} \cdot \mathbf{N}|_{\partial \mathcal{M}_{out}} \quad (5)$$

in the muscle layer; here, δR_o is the thickness of the muscle layer in the reference configuration, $\mathbf{N}$ is the outward unit normal to the cylinder, and the pressure exterior to the cylinder has been set to zero. The operator $\overline{\text{Div}}$ is the usual three-dimensional divergence, while $\overline{\text{Div}}$ is the surface divergence (i.e., a three-dimensional divergence whose radial terms have been projected away). Thus, our thin muscle layer approximation has reduced a set of coupled partial differential equations on multiple domains to a single-domain system with an effective boundary condition on the outer surface of the cylinder [49]. The boundary conditions on the remaining surfaces are

$$-P_{in} \mathbf{F}^{-T} \mathbf{N} = \mathbf{P} \cdot \mathbf{N}|_{\partial \mathcal{M}_{in}} \quad (6)$$

$$\mathcal{P}_{rZ}|_{\partial \mathcal{M}_{top/bot}} = \mathcal{P}_{\theta Z}|_{\partial \mathcal{M}_{top/bot}} = 0 \quad (7)$$

$$z|_{\partial \mathcal{M}_{top/bot}} = \pm L/2 \text{ or } \mathcal{P}_{zZ}|_{\partial \mathcal{M}_{top/bot}} = 0, \quad (8)$$

where $z(R, \Theta, Z)$ is the post-deformation axial coordinate, P_{in} is the pressure in the lumen and L is the reference esophagus length. We also impose that the deformation is continuous at the interface. There are two complementary boundary conditions that we can impose at the cylinder ends, given by the two alternatives in Eq. (8): the axial or “Dirichlet” boundary condition (first alternative) or the free boundary or “Neumann” condition (second alternative). In the axial boundary condition, the axial stretch ratio λ_z , defined as the post-deformation cylinder length divided by the reference length, is prescribed, and thus in general an external axial force F_z must be applied at both ends to maintain equilibrium (the shear stresses at the ends are still assumed to vanish). Balancing the forces in the Z -direction at the top face of the cylinder in the deformed configuration informs us that

$$F_z \mathbf{E}_Z = \frac{2\pi}{\lambda_z} \left(R_o \delta R_o \bar{\mathbf{P}} \bar{\mathbf{F}} + \int_{R_i}^{R_o} dR R \mathbf{P} \mathbf{F} \right) \cdot \mathbf{E}_Z. \quad (9)$$

Since the esophagus is mechanically anchored to the pharynx proximally and the stomach distally, its overall length is constrained by the surrounding anatomy. As a result, axial deformation is restricted by available spatial confinement rather than determined solely by internal forces. To capture this physiological constraint, we set the initial and final lengths to be the same, so that the axial stretch parameter $\lambda_z = 1$ when using the axial boundary condition.

We will also find it useful to consider the complementary free boundary problem, in which the cylinder is free to change length and the stress vanishes at the top and

bottom surfaces. Strictly speaking, it is not possible to satisfy $\mathbf{P} \cdot \mathbf{N} = \mathbf{0}$ exactly when λ_z is constant, as p , and therefore $\mathcal{P}_{zZ} = \mu \lambda_z - (p/\lambda_z)$, depends on R . Thus, in lieu of setting $\mathcal{P}_{zZ}$ to zero, the external force F_z is set to zero and the no-stress boundary condition is satisfied in an averaged sense instead [46]. This change is not expected to meaningfully affect our results, as Saint Venant’s principle assures us that the deformation fields corresponding to different boundary load distributions have little impact away from the boundary [50], and the esophagus has a high aspect ratio. The additional condition $F_z = 0$ then allows the axial deformation λ_z to be determined. Solving the problem with this boundary condition will allow us to gauge the extent of lengthening of the esophagus in response to muscle contractions, a phenomenon that is observed in our patients with corkscrew esophagus. In reality, the ends of the esophagus are neither completely constrained nor completely free but move with some resistance. Thus, our two complementary sets of boundary conditions define a range of deformations and provide theoretical lower and upper bounds for any model outputs for a given set of input parameters.

D. Dilation, Contraction, and Axial Force

To build mechanical intuition, we first analyze the case in which the esophagus deforms axisymmetrically and in a manner independent of Z , which is a modification of the classic inflation–extension of a tube [16, 46, 51–54]. Torsion is also possible if applied shear stresses are present [55, 56], which can happen if $\mathbf{M}_1$ and $\mathbf{M}_2$ are varied, but we ignore these effects for simplicity. (An analysis of the effect of torsion appears in the Supporting Information; the addition of torsion does not significantly alter our results.) Under these assumptions, the active hoop stress simply contributes a radial force, similar to a surface tension, causing the cylinder to contract. We define $r_i = r(R_i)$ and $r_o = r(R_o)$ to be the corresponding radii of the deformed cylinder. The incompressibility and axisymmetry of the body demand that $r(R) = \sqrt{(R^2 + c)/\lambda_z}$ for some constant c . Plugging this into Eq. (4) reveals that

$$p = \frac{\mu}{2\lambda_z} \left(\frac{R^2}{R^2 + c} - \log \frac{R^2}{R^2 + c} \right) + q, \quad (10)$$

where q is a constant of integration. The two unknowns c and q can be determined by applying the boundary conditions Eqs. (5) and (6); in this way, the dilation of the esophagus depends on the lumen pressure P_{in} as well as the sheet parameters.

Wide ranges of esophageal mechanical parameter values have been reported in the literature, many of which are specific to certain models, constitutive laws, or even patients. For example, reported esophageal wall thickness values vary with both imaging modality and the anatomical layer measured. On one hand, intraluminal ultrasound studies show that type III achalasia patients

have thicker muscle layers than normal subjects, with a CM thickness of 1.71–2.33 mm and an LM thickness of 0.85–0.93 mm, compared with approximately 0.68 mm (CM) and 0.66 mm (LM) in healthy individuals [32]. On the other hand, multidetector computed tomography studies similarly report increased overall wall thickness in type III achalasia patients, with mean values of about 6 mm in the esophageal body and 10 mm at the distal esophageal sphincter [57]. Based on these considerations, we take the undeformed radii of the cylinder to be $R_i = 3$ mm and $R_o = 10$ mm and the muscle layer thickness to be $\delta R_o = 2$ mm in our analysis. The elastic moduli for the various esophageal layers are also debated, and no clear consensus has emerged. Here, we set the elastic modulus of the muscle matrix to $\bar{\mu} = 2$ kPa. We also set the elastic modulus of the mucosa to be $\mu = 1.4$ kPa. While this is approximately an order of magnitude less than reported values that consider the resistance of embedded fibers [58], we expect the mucosa to be in a compressed state and thus effectively much easier to deform than normal. Estimates of the fiber elastic moduli $\bar{\mu}_1$ and $\bar{\mu}_2$ are extremely limited in the literature and range approximately from 1 to 100 kPa [47, 59], with $\bar{\mu}_1$ known to be less than $\bar{\mu}_2$. Based on an order-of-magnitude comparison of the terms in our fiber strain energy function Eq. (3) and the leading order elastic terms of the model in [59], we set $\bar{\mu}_1 = 40$ kPa for the circumferential fibers and $\bar{\mu}_2 = 80$ kPa for the longitudinal fibers. As will be demonstrated, varying these parameters within an order of magnitude tunes our numerical predictions but does not impact the general physical mechanism that leads to instability.

Using the above parameters, the pressure–dilation relationship of three different elastic cylinders is shown in Fig. 3A. The leftmost curve (light blue) is for a cylinder with no elastic sheet, for which classical theory predicts that the dilation diverges at a finite critical pressure of $P_{in}^* = (\mu/\lambda_z) \log(R_o/R_i)$ [16]. Incorporating the active circumferentially contracting fiber layer described above modifies this rupture pressure to

$$P_{in}^* = \frac{\mu}{\lambda_z} \log \frac{R_o}{R_i} + \frac{\delta R_o (\bar{\mu} + \lambda_1^2 \bar{\mu}_1)}{\lambda_z R_o}, \quad (11)$$

which is shown by the other curves in the figure. For parameters representative of the typical human esophagus, the sheet contribution to Eq. (11) is an order of magnitude greater than the bulk contribution. Thus, the additional contractile hoop stress allows the channel to sustain substantially higher internal pressures at comparable radii; indeed, some of the very high pressures observed in clinical manometry data are several times greater than the classical rupture pressure and are only possible with active CM contraction and values of $\bar{\mu}_1$ on the order of tens of kilopascals. The cross-sectional area is also larger when contractile forces are present (Fig. 3A inset), a consequence of the conservation of mass.

Figures 3B and C show intraluminal pressure as a function of inner radius and CM contraction strength under

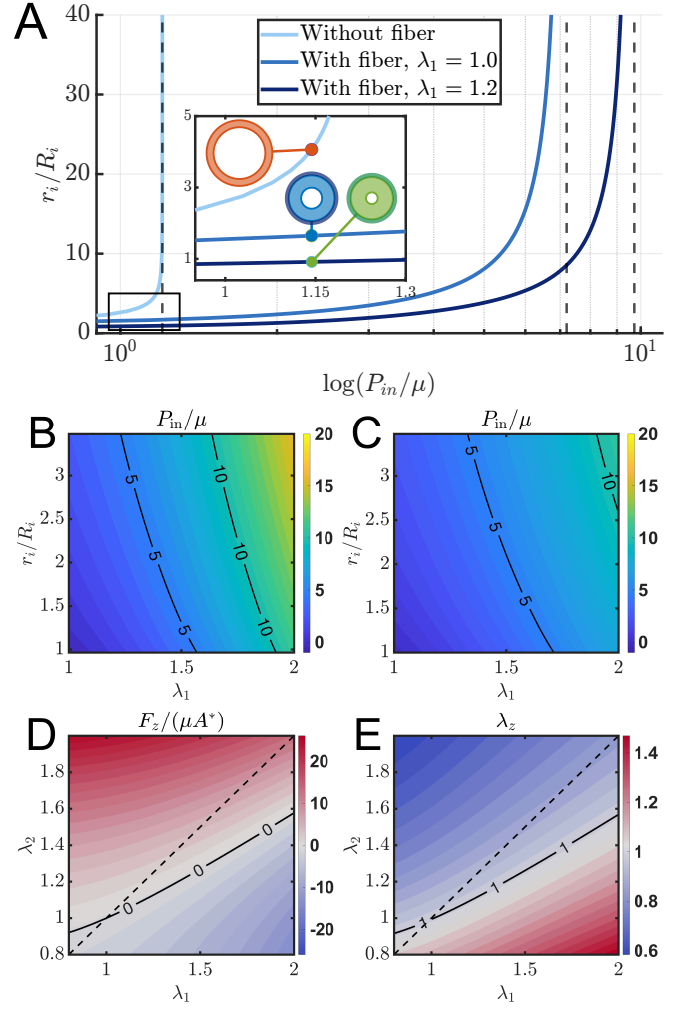


FIG. 3: Geometry of the tube in response to internal pressure and active contraction. Parameters are $\mu = 1.4$ kPa, $\bar{\mu} = 2$ kPa, $\bar{\mu}_1 = 40$ kPa, $\bar{\mu}_2 = 80$ kPa, $R_i = 3$ mm, $R_o = 10$ mm, and $\delta R_o = 2$ mm. The axial stretch ratio $\lambda_z = 1$ for the fixed boundary condition and is determined using $F_z = 0$ for the free boundary condition. (A) Critical pressure (asymptote) of an elastic tube with and without fibers and active circumferential contraction under fixed boundary conditions; inset, deformed cross-section at a representative pressure and contraction state. (B,C) Inner pressure dependence on inner radius r_i/R_i and circumferential contraction λ_1 under (B) fixed axial and (C) free boundary conditions. (D) Dimensionless axial force $F_z/(\mu A^*)$ as a function of circumferential (λ_1) and longitudinal (λ_2) contraction under fixed boundary conditions. Here, $A^* = \pi(R_o^2 - R_i^2)$. (E) Axial stretch λ_z as a function of circumferential and longitudinal contraction under free boundary conditions. In (D) and (E), the dashed line indicates where $\lambda_1 = \lambda_2$.

fixed and free axial boundary conditions, respectively. Larger λ_1 and larger inner r_i/R_i both lead to higher P_{in}/μ , reflecting the force balance between active circumferential tension, wall stresses, and internal pressure at equilibrium. When P_{in} is held constant, increasing λ_1

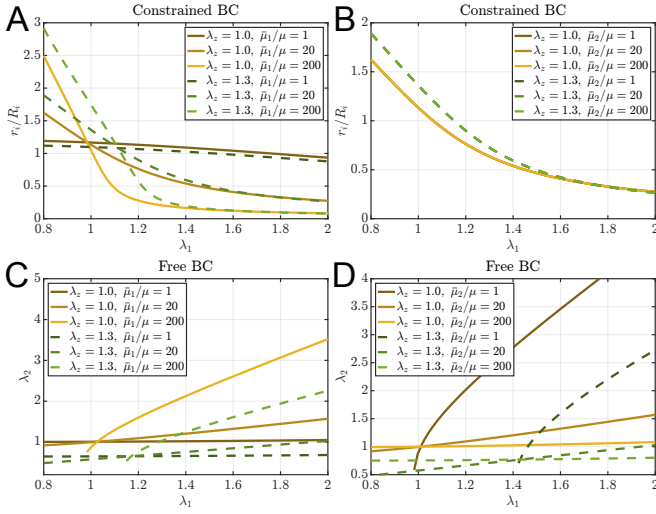


FIG. 4: Impact of fiber stiffness and axial stretch on geometry and contraction. The stiffness of the orthogonal fiber family is fixed at $\bar{\mu}_i/\mu = 20$ ($i = 1, 2$). (A,B) Normalized inner radius r_i/R_i versus circular contraction under fixed boundary conditions at $P_{in}/\mu = 0.267$ for varying circular (A) and longitudinal (B) muscle stiffness at different axial stretches. (C,D) Circular versus longitudinal contraction under free boundary conditions for varying circular (C) and longitudinal (D) muscle stiffness at different axial stretches.

causes the inner radius to decrease, reflecting hoop shortening regardless of which boundary condition is applied at the cylinder end. In accordance with incompressibility, a radially inward directed compression causes axial lengthening of the cylinder. If the ends of the cylinder are held fixed, then this lengthening instead manifests itself as a compressive force (negative F_z) of a greater magnitude (Fig. 3D). If instead the cylinder is allowed to freely elongate, the resulting λ_z exceeds 1 whenever the circumferential contraction is sufficiently strong relative to the longitudinal contraction, i.e., for all (λ_1, λ_2) below the $\lambda_z = 1$ contour in Fig. 3E. Figs. 3D–E also show that increasing λ_2 has the reverse effect: the esophagus will tend to shorten (λ_z decreases), and if the ends are constrained to remain fixed, increasingly tensile forces must be applied. Notably, under a free axial boundary condition ($F_z = 0$), axial lengthening occurs only when circumferential contraction dominates longitudinal shortening. When circumferential and longitudinal activation are equal ($\lambda_1 = \lambda_2$, Fig. 3D–E, dashed line) and greater than or equal to 1, the free solution satisfies $\lambda_z \leq 1$, indicating no intrinsic tendency toward axial extension; in fact, the esophagus is predicted to shorten.

Figure 4 shows how contractility and fiber stiffness regulate luminal geometry and the coupling between circular and longitudinal contractions. Increasing circumferential contraction consistently decreases the inner radius, again reflecting hoop shortening under incompressibility. By contrast, the effect of axial stretch λ_z is not monotonic: in Fig. 4A, the inner radius increases with axial stretch

at low circumferential stiffness but decreases with axial stretch at high circumferential stiffness. This behavior reflects two competing volume-preserving responses to axial elongation, namely radial expansion with wall thinning and radial contraction with wall thickening. Increasing circumferential stiffness shifts the balance toward lumen collapse, as the stiff outer muscle layer acts as a contracting shell. In contrast, Fig. 4B shows that longitudinal muscle stiffness has no effect on the luminal geometry, as $\bar{\mu}_2$ does not enter into Eq. (5) at all.

Under free boundary conditions (Fig. 4C–D), circular and longitudinal contractions are strongly coupled. Fig. 4C shows that stronger circumferential contraction requires stronger longitudinal contraction, and that this coupling increases with circumferential stiffness when longitudinal stiffness is fixed. Increasing axial stretch reduces the longitudinal contraction required for a given circumferential contraction, consistent with axial pre-tension partially accommodating circular shortening. Fig. 4D further shows that, for fixed circumferential stiffness, smaller longitudinal stiffness requires larger longitudinal contraction to maintain the same circumferential contraction. Together, these results provide a sensitivity analysis of fiber stiffness and help identify plausible parameter ranges for comparison with clinical data.

These trends are highly consistent with the findings of experimental and modeling studies indicating that CM contraction primarily contributes to intraluminal pressure generation through hoop stress balance, whereas longitudinal muscle contraction is associated with wall thickening and axial force [60]. An important clinical feature of corkscrew esophagus is near-simultaneous contraction along the esophageal body. However, HRM data do not demonstrate substantial upward displacement of the lower esophageal sphincter (LES), suggesting that there is limited net axial translation during these swallows. It is notable that axial shortening is sometimes observed 10 to 30 s *after* swallowing is completed, which is discussed below. Because the axial distance between the upper and lower esophageal sphincters remains approximately fixed while the deformed configuration becomes helical, the arc length of the esophageal centerline must increase relative to its projected length. This observation motivates the hypothesis that strong concurrent circumferential contraction alters the preferred axial stretch. If axial displacement is constrained, a mismatch between the preferred and imposed lengths may generate axial compression and trigger buckling, analogous to how increased length and compressive force make a slender rod or elastica more susceptible to buckling [61].

E. Estimation of Contractility from High-Resolution Manometry Data

Using the method described in [26], we calculate the lumen radius at each time and sensor from the impedance data (Fig. 1C and G). To quantify the contraction

strengths λ_1 and λ_2 , we analyze a two-stage deformation with two different axial boundary conditions. In the first stage, the manometer is placed inside of the esophagus, and we solve for λ_1 using the manometry pressure during contraction and the constrained boundary condition ($\lambda_z = 1$). During manometry trials for patients with very high contractility (typically those with achalasia III), the esophageal wall comes into contact with the manometer itself. This is confirmed by the impedance data, which show that the cross-sectional area approaches a lower bound (Fig. 1C), as well as the fact that the measured contact pressures for achalasia III patients are often significantly larger than baseline lumen pressures. Since the radius of the manometer is known (2.1 mm), the measured contact pressure can be used as P_{in} in Eq. (6) and combined with Eq. (5) to calculate λ_1 (Fig. 1D and H). For our analysis, we manually select a rectangular box in the time–sensor plane around the region of peak pressure of each swallow in the HRM data (examples are shown in Fig. 1B and F), and the state of that swallow is defined as the mean of P_{in} , and hence of λ_1 , over the sensors and the duration of the box (see Methods). This value is most accurate when the pressure variation in the vertical direction is small, consistent with the assumption of axisymmetry and independence of Z .

In the second stage, the manometer has been removed and the esophagus may have buckled and elongated after being destabilized. We suppose that the esophagus is “straightened out” by removing the boundary constraints so that the ends are force-free but its contraction parameters λ_1 and λ_2 have not changed. The axial stretch λ_z is measured from the esophagrams as the arc length of the esophageal centerline during the swallow (a helix in a corkscrew patient) divided by the length of the centerline at rest, when it is a straight line from the top to the bottom of the esophagus (see Methods). Assuming the same radial deformation and pressure from the first stage, we can find λ_2 using the axial force balance condition Eq. (9) with the measured λ_z and with F_z set to 0. Thus, we have a procedure to infer the contractility parameters λ_1 and λ_2 from clinical data. Note that when the fibers are perfectly circumferentially and longitudinally oriented, i.e., $\mathbf{M}_1 = \mathbf{E}_\Theta$ and $\mathbf{M}_2 = \mathbf{E}_Z$, the radial boundary condition involves λ_1 but is independent of λ_2 , simplifying the equations.

The calculated λ_1 and λ_z values of each patient are shown in Fig. 5. Taking noncorkscrew patients to have $\tilde{k} = 0$, we see from Fig. 5A that λ_z correlates positively with the dimensionless wavenumber $\tilde{k}$. However, no analogous trend is seen for λ_1 (Fig. 5B). Figures 5C and D emphasize this point: using Student’s t -test, we find that the difference in λ_z between the corkscrew and noncorkscrew groups is statistically significant, while the difference in λ_1 is not. Together, these figures strongly suggest that longitudinal contraction plays a greater role in corkscrew formation compared with circumferential contraction. While λ_1 is not sufficient to differentiate between corkscrew and noncorkscrew achalasia III pa-

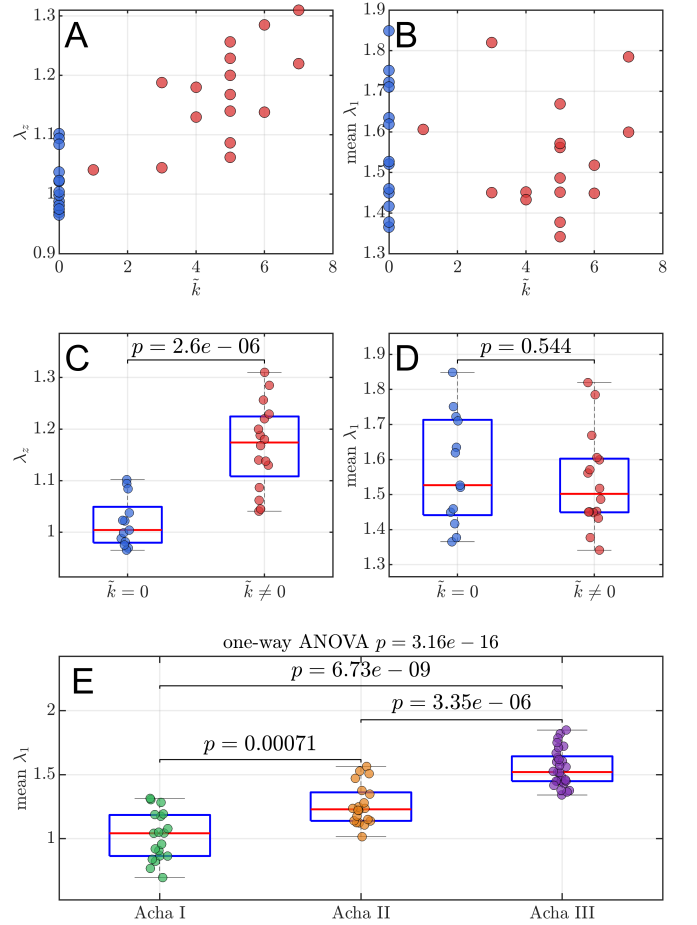


FIG. 5: Clinical statistics of inferred esophageal parameters across patient groups. (A) Axial stretch ratio λ_z as a function of dimensionless wavenumber $\tilde{k} = kL/\pi$ for both corkscrew ($\tilde{k} \neq 0$, red) and non-corkscrew ($\tilde{k} = 0$, blue) achalasia III patients. (B) Clinically observed circumferential contractility parameter λ_1 for the patients in (A). The mean λ_1 for each patient is computed by averaging over all of that patient’s swallows. (C) A statistically significant difference is observed in the axial stretch λ_z between the corkscrew and non-corkscrew patients in (A). (D) No significant difference is detected in λ_1 for the patients in (A). (E) Calculated circumferential contractility λ_1 for different classes of achalasia patients. The p-values indicate that the differences between groups are statistically significant.

tients, it can differentiate among the different types of achalasia (Fig. 5E), which is consistent with clinical evidence and definitions. Figure 6 shows that the reference lumen radius is significantly different both among different achalasia types (Fig. 6A) and between corkscrew and noncorkscrew patients (Fig. 6B). We discuss the implications of these trends in Sec. III.

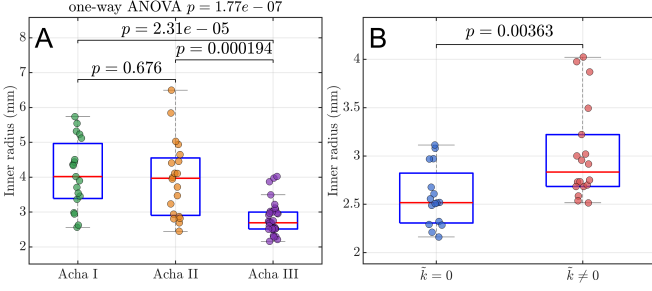


FIG. 6: Uncontracted inner radius measurements for achalasia patients derived from HRM data. (A) Comparison of inner radius among patients with type I, type II, and type III achalasia. (B) Comparison of inner radius within type III achalasia patients, separated by whether a nonzero k -mode is present. Measurements were obtained while the esophagus was at rest, using the best-estimated resting segment at the beginning or end of the recording. The inner radius for each case was defined as the mean value across sensors 10–30.

F. Continuum Model Predicts Instability of the Esophagus

Clinically, corkscrew esophagus is characterized by concurrent esophageal contractions over a long segment of the esophagus [38]. We hypothesize that these concurrent contractions induce a mechanical instability within the esophageal wall, leading to a change from a relatively cylindrical shape to a buckled or corkscrewed morphology. In analogy with the buckling of a slender rod [51], we expect a critical contraction strength above which the straight configuration loses stability. Specifically, we focus on how increasing CM or LM contractility influences the system's susceptibility to buckling.

To analyze the stability of the axisymmetric configuration studied above, we add a perturbation $\delta\mathbf{x}$ to the deformation field and a perturbation δp to the pressure field, both with dimensionless amplitudes of order $\epsilon \ll 1$ relative to their respective base case quantities. We also enforce the condition of fixed L^2 norm of $\delta\mathbf{x}$ [62]. The corresponding perturbation to the deformation gradient is $\delta\mathbf{F} = \nabla\delta\mathbf{x}$, and $\delta\mathbf{P} = \mu\delta\mathbf{F} - \delta\mathbf{F}^{-T} + p\mathbf{F}^{-T}\delta\mathbf{F}\mathbf{F}^{-T}$ is the resulting perturbation to the first Piola–Kirchhoff stress. Expanding the total strain energy $W_e + \bar{W}_e + \bar{W}_a$ to second order in ϵ yields an eigenvalue problem

$$\text{Div}(\delta\mathbf{P} + \zeta\delta\mathbf{F}) = \mathbf{0} \quad (12)$$

with incompressibility constraint

$$\text{tr}(\mathbf{F}^{-1}\delta\mathbf{F}) = 0 \quad (13)$$

and eigenvalue ζ . The associated boundary conditions for a constrained esophagus are

$$(\delta\mathbf{P} + \zeta\delta\mathbf{F}) \cdot \mathbf{N} = -P_{in}\delta\mathbf{F}^{-T} \cdot \mathbf{N}|_{\partial\mathcal{M}_{in}} \quad (14)$$

$$(\delta\mathbf{P} + \zeta\delta\mathbf{F}) \cdot \mathbf{N} = \delta R_o \overline{\text{Div}}(\delta\bar{\mathbf{P}} + \zeta\delta\bar{\mathbf{F}})|_{\partial\mathcal{M}_{out}} \quad (15)$$

$$\begin{aligned} \delta z|_{\partial\mathcal{M}_{top/bot}} &= \delta\mathcal{P}_{rZ} + \zeta\delta F_{rZ}|_{\partial\mathcal{M}_{top/bot}} \\ &= \delta\mathcal{P}_{\theta Z} + \zeta\delta F_{\theta Z}|_{\partial\mathcal{M}_{top/bot}} = 0 \end{aligned} \quad (16)$$

In this formulation, the sign of the smallest eigenvalue ζ determines the stability of the axisymmetric configuration (the second variation operator is known to be Hermitian and thus the eigenvalues are guaranteed to be real [63]). We Fourier expand the perturbations: $\delta\mathbf{x} = \sum_{m,k} \delta\mathbf{x}_{mk} e^{im\Theta} e^{ikZ}$ and $\delta p = \sum_{m,k} \delta p_{mk} e^{im\Theta} e^{ikZ}$, where the coefficients $\delta\mathbf{x}_{mk}$ and δp_{mk} depend only on R . Here, the azimuthal mode number m is an integer because of the periodicity of Θ , and $\tilde{k} = kL/\pi$ must also be an integer to satisfy the boundary conditions at the ends. Since small-amplitude corkscrew perturbations resemble buckling modes [64], and since it is known that long, slender structures are generally more susceptible to buckling than to barreling instabilities [51], we focus on the case $m = 1$. We show in the Supporting Information that the other m modes are less unstable for the range of k that we consider here. The generalized eigenvalue problem given by Eqs. (12)–(16) is solved using a finite element method [65]. A stability diagram plotting the critical contractility λ_1 needed to destabilize the cylinder as a function of dimensionless wave number $\tilde{k}$ is shown in Fig. 7a. Representative solutions are shown in Fig. 7b. In general, the most unstable mode is $\tilde{k} = 1$, and the λ_1 necessary to destabilize the tube increases with $\tilde{k}$. Increasing the longitudinal contraction λ_2 is seen to have the inverse effect of stabilizing the system. This is due to the fixed boundary conditions, which necessitate applied tensile forces at the boundary, lowering the overall compressive stress in the esophageal wall.

Clinical observations indicate that corkscrew esophagus presents with a range of spatial wavenumbers, $1 \leq k \leq 7$, most commonly $k = 5$. In our analysis, the most unstable mode, defined as the $\tilde{k}$ whose corresponding ζ is the smallest, is always $\tilde{k} = 1$. This is consistent with the spectrum of a buckling rod: the lowest mode requires the smallest compressive force, and increasingly higher modes require increasingly larger loads. The linear analysis therefore predicts the *onset* of instability. The observed $\tilde{k} > 1$ is expected to be selected in the post-buckling regime, where the finite amplitude of the helix, the lateral tethering of the esophagus to surrounding mediastinal tissue (which acts as an elastic foundation and penalizes long-wavelength lateral excursions), and the finite length of the spastic segment all favor shorter wavelengths than the free $\tilde{k} = 1$ Euler mode. Resolving this wavelength selection requires a nonlinear post-buckling analysis that is beyond the scope of the present work. Sensitivity analysis of the perturbation solution indicates that the model's buckling behavior is not highly sensitive to typical physiological pressure fluctuations, implying that P_{in} may not be the primary determinant of corkscrew morphology. This is consistent with clinical data suggesting that intraluminal pressure variations among patients are relatively modest [66]. A diagram illustrating stable and unstable regions in the

λ_2 - λ_1 -plane is shown in Fig. 8. The marginal stability curve that separates the unstable and stable regions is plotted assuming that $m = \tilde{k} = 1$ (since this is the most unstable mode) and $P_{in} = 0.8$ kPa, a representative resting lumen pressure below 10 mmHg.

Because the swallows of a given patient are not independent, we report classification metrics at the patient level ($n = 29$) rather than at the swallow level ($n = 278$). A patient is predicted to corkscrew if the patient-mean state lies on the unstable side of the curve (Fig. 8A); Fig. 8B shows that this prediction is robust in the sense that, for nearly every patient, either all or none of the individual swallows lie past the curve. Every patient with an observed helix lies past the boundary (recall = 1.00, 16/16), while 9 of the 13 patients without a helix lie on the stable side (specificity = 0.69; precision = 0.80, accuracy = 0.86). The receiver operating characteristic in Fig. 8C, obtained by thresholding each patient's mean signed distance from the curve, has an area under the curve of 0.95. Notably, the four false positives are precisely the $\tilde{k} = 0$ patients with the largest measured λ_z (1.04–1.10), i.e., those closest to the corkscrew group in Fig. 5A, which suggests that they are near-threshold cases rather than failures of the mechanism. The color of the data points in Fig. 8 represents the patient's λ_z estimated from esophagrams. There is a clear negative correlation between λ_2 and λ_z as weaker axial contraction allows for more elongation of the esophagus.

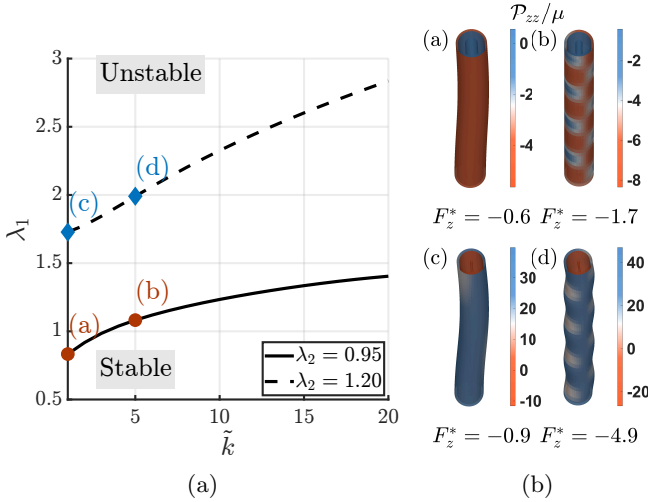


FIG. 7: An elastic tube experiences instability under sufficiently strong circular contractions and compressive axial forces. (a) Stability diagram of circumferential contraction as a function of wave number $\tilde{k}$ for different λ_2 . The labeled points on the curves correspond to the shapes shown in (b). In (b), color indicates the perturbed first Piola-Kirchhoff stress distribution on the mucosa layer and on the active surface, and F_z^* is the dimensionless total axial force of the tube. Negative values of F_z^* represent compressive forces. Different amplitudes were chosen for different cylinders for visual clarity.

III. DISCUSSION AND CONCLUSIONS

A. Longitudinal and Circumferential Muscle Incoordination

The fact that corkscrew esophagus is not observed in patients with type II or type I achalasia, whereas it can sometimes be seen in patients with hypercontractile esophagus, would strongly suggest that the driver of instability is abnormally forceful muscle contractions [38]. As we have seen, increasing λ_1 leads to a tendency of the esophagus to lengthen, which seemingly could explain the larger λ_z found in our corkscrew patients (Fig. 5A), and it is thus natural to hypothesize that λ_1 should be larger for such patients. Yet, this hypothesis is directly contradicted by Fig. 5D, which shows that no statistically significant difference in λ_1 between corkscrew and non-corkscrew patients exists ($p = 0.544$). Even when λ_1 is calculated using force-free boundary conditions to allow for the effect of axial lengthening, this difference is not significant ($p = 0.402$), and the increased λ_1 alone is not sufficient to explain this discrepancy. Conversely, Fig. 8 shows that corkscrew patients tend to have lower λ_2 values (which is consistent with the higher λ_z values shown in Fig. 5). The reference inner radius does differ between the two groups (Fig. 6B), but, as shown below (Fig. 9A), varying R_i over the observed range shifts the marginal stability curve only modestly, so this geometric difference cannot by itself account for the separation of the two populations. Patient-specific elastic moduli are not available, and we assume that they do not differ systematically between the groups. We thus believe that a weaker longitudinal contraction strength λ_2 is more likely responsible for the corkscrew phenomenon during swallowing, signaling LM-CM incoordination or possibly LM impairment more generally.

There is considerable evidence for the phenomenon of LM-CM incoordination in patients with esophageal motility disorders. In healthy individuals, the two muscle layers are known to contract in a highly synchronized manner during deglutition [67]—the fine coordination between the two muscle layers reduces wall stress and makes bolus transport substantially more efficient. Conversely, LM-CM temporal asynchrony has been directly demonstrated in spastic motility disorders such as nutcracker esophagus [68], and incoordination between layers has been discussed in the context of achalasia and other disorders [29, 69]: specifically, in type III achalasia, the circular and longitudinal muscles both contract but show severe incoordination. Delayed LM contraction after performing a rapid drink test has also been observed in achalasia patients [70]. Our clinical manometry data display similar strong longitudinal contractions approximately 10 to 30 s after swallowing has concluded, as evidenced by the vertical displacement of the high-pressure region demarcating the EGJ. Together, these trends lend credence to the idea that by increasing the length of the esophagus, decreasing its effective stiffness, and adding compressive

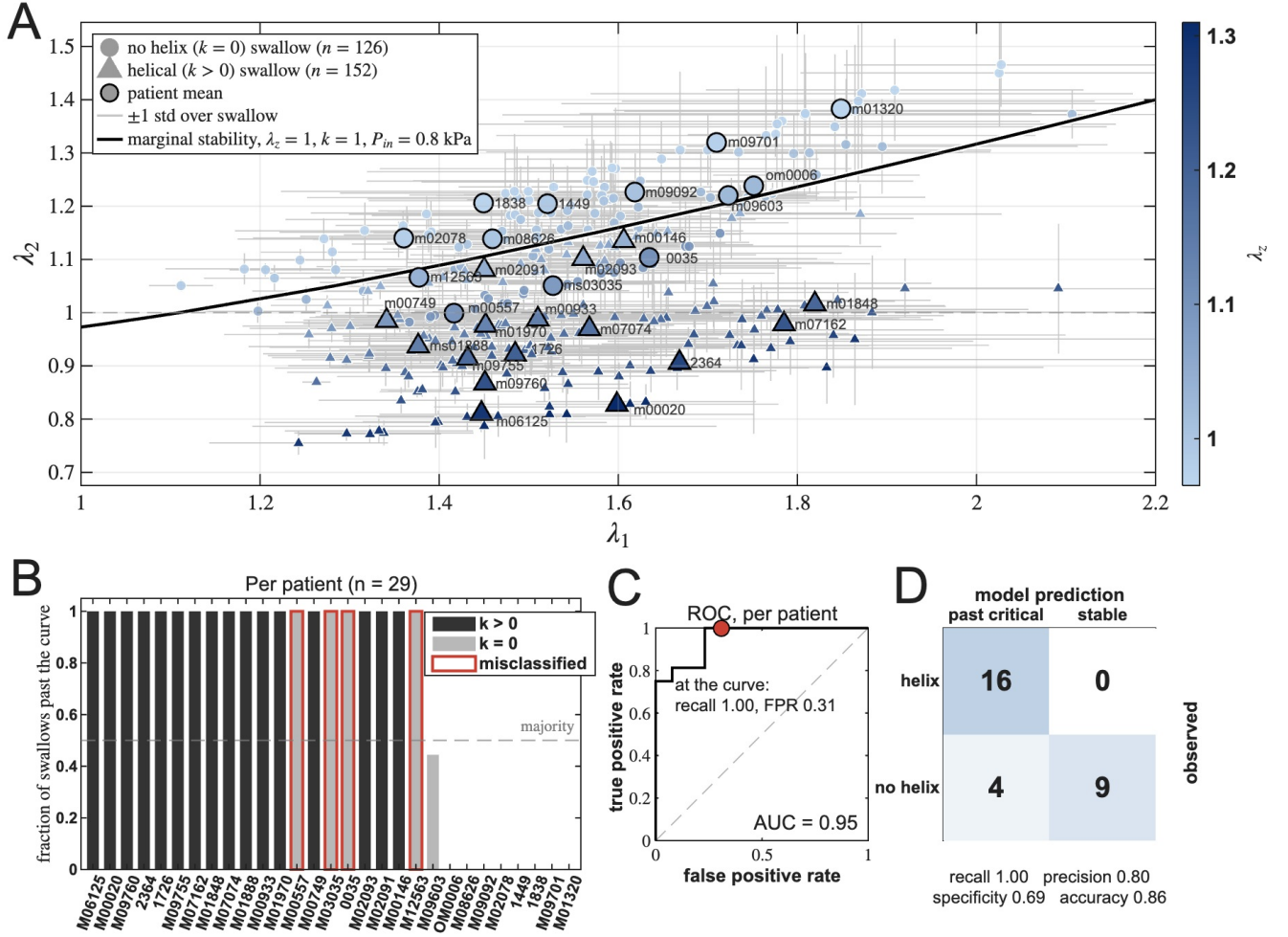


FIG. 8: Marginal stability of the $m = \bar{k} = 1$ mode versus clinical data from 29 patients with type III achalasia. (A) Swallow-averaged (λ_1, λ_2) for 278 swallows (small markers; grey bars show ± 1 s.d. of the sensor-averaged values over the swallow duration) and patient means (large markers), colored by λ_z from patient esophagrams. Triangles: observed helical deformation ($\bar{k} \neq 0$, $n = 16$); circles: none ($\bar{k} = 0$, $n = 13$). Black curve: marginal stability boundary at the pre-instability state ($\lambda_z = 1$, $P_{in} = 0.8$ kPa); states to its right are unstable. (B) Fraction of each patient's swallows past the curve; red outlines mark the four $\bar{k} = 0$ patients whose mean state lies on the unstable side. (C) Receiver operating characteristic obtained by thresholding the per-patient mean signed distance to the curve (area under curve = 0.95); the red point marks the operating point corresponding to the curve itself (threshold zero). (D) Confusion matrix at that operating point: recall 1.00, precision 0.80, specificity 0.69, accuracy 0.86.

stresses, LM dysfunction plays a central role in corkscrew esophagus and possibly in esophageal motility disorders more broadly.

B. Esophageal Wall Remodeling and Disease Progression

Our model also points to additional biomechanical factors that may be involved in the progression of type III achalasia into more severe forms of the disease. Clinically, type I and type II achalasia are often associated with greater dilation and diminished contractility compared with type III [71]. As demonstrated in Fig. 5E,

there is a clear progressive decline in circumferential contractility across the three achalasia subtypes (when viewed in the order III, then II, then I) for the patients in our dataset. Furthermore, Fig. 6 indicates that the inner lumen in the esophagus's uncontracted state is markedly more dilated in type I and type II achalasia than in type III. Taken together, these trends are consistent with the conjecture that the three subtypes of achalasia defined in the Chicago classification represent different stages of the same disease [35, 36]: when viewed in the order of achalasia type III, then II, then I, the progression implies that the esophagus initially has overly strong contractions but gradually loses its ability to contract while it simultaneously becomes looser and more dilated. Such

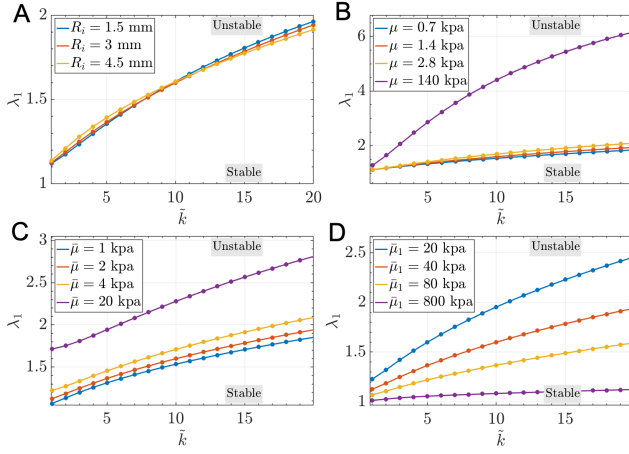


FIG. 9: Stability diagrams of λ_1 as a function of wavenumber under different esophageal body properties. (A) Effect of varying the reference inner radius R_i (B) Effect of varying mucosal stiffness μ . (C) Effect of varying muscle matrix stiffness $\bar{\mu}$. (D) Effect of varying muscle fiber stiffness $\bar{\mu}_1$.

drastic physiological changes are thought to be related to the phenomenon of strain softening or tissue remodeling, in which repeated large deformations reduce the effective elastic modulus of the esophageal tissue [72].

Figure 6 also shows that corkscrew patients display a similar trend of larger luminal radius, with an average radius less than that of type II and I achalasia patients but still significantly greater than that of non-corkscrew patients. As shown in Fig. 9A, variation of the reference inner radius R_i alone has only a modest direct effect on the marginal stability curve. In contrast, material properties have a much stronger influence: decreasing the mucosal stiffness μ or passive muscle matrix stiffness $\bar{\mu}$ clearly destabilizes the system (Fig. 9B and C), whereas decreasing the active CM fiber stiffness $\bar{\mu}_1$ stabilizes it (Fig. 9D). These results suggest that while a larger inner radius may not itself be the primary mechanical cause of instability, it may rather be a geometric marker of underlying wall remodeling that has altered the effective stiffness of the esophageal body. In particular, lumen dilation may reflect progressive softening of the mucosa and passive matrix caused by the strong panesophageal spasms characteristic of corkscrew esophagus, making the wall more compliant, while changes in the active muscle response may further shift the system toward instability. From this perspective, corkscrew esophagus may share part of the same remodeling trajectory, while remaining distinguished by sufficient residual or unbalanced active contraction to drive a non-axisymmetric instability.

C. Limitations of the Study

While we are able to capture much of the physics of corkscrew esophagus with our simple minimal model and identify a necessary condition for its occurrence in patients, we must acknowledge certain assumptions that were made for expediency or parsimony and may not be realistic. First and foremost, our model assumes a simplified, largely uniform esophagus. In reality, the composition of the esophagus is highly heterogeneous and includes more than just two types of tissue. A large knowledge deficit regarding the constitutive laws and parameters that govern each of these layers exists [39, 58, 59]. Axially, the esophagus is not uniform—the lower, middle, and upper sections of the esophagus have different muscle compositions, have varying muscle fiber angle orientations, and are connected to different parts of the gastrointestinal tract [73]. These properties will potentially alter the mechanics of each region, and models that account for this are expected to perform better.

Our model also does not take into account dynamical phenomena or explicitly model the bolus or the obstruction at the EGJ and thus provides limited information about the actuation of corkscrewing or the postbuckled state. For simplicity, the contraction strengths λ_1 and λ_2 were taken to be constant in this study. In reality, spatial and temporal fluctuations are likely present. Such fluctuations could explain the transient, high-wavenumber tertiary contractions that are present even in healthy patients [74]. On longer timescales, muscle hypertrophy and atrophy have been proposed to play roles in the evolution of corkscrew esophagus and achalasia over time [29]. These effects may violate the thin membrane assumption used to simplify the mechanics of the muscle layer.

High-resolution manometry is an effective method for studying CM function of the esophagus but provides little direct, quantitative information on the LM, given its difficulty in measuring axial shortening and lengthening of the esophagus [29, 60]. Our two-stage procedure at best produces a rough estimate for λ_2 . Quantifying the axial force F_z would help more accurately determine λ_2 and provide a more direct confirmation of the hypothesized LM-CM incoordination. Patient-specific parameters (e.g., elastic moduli) were also not available for this study. Additional studies that make use of tools capable of gathering such information, such as functional lumen imaging probe (FLIP) manometry [75], could be useful for further parsing the various effects that contribute to corkscrewing of the esophagus.

D. Conclusion

We have proposed an intuitive mechanism for the corkscrew phenomenon in terms of elastic instability and confirmed it with a minimal mathematical model and clinical data analysis. The corkscrew-like deformation can be seen as an instance of a hyperelastic instability in-

stigated by strong internal compressive stresses brought on by active circumferential contractions. At the same time, the longitudinal muscle is impaired, exacerbating the effects of CM contraction. Similar to a compressed rod, the esophagus then buckles once a critical contractility is exceeded; the number of turns that develops is set in the post-buckling regime and is not predicted by the linear theory. No parameter in our theory was fitted to the clinical outcome: all material and geometric values were fixed from the literature before comparison with the patient data.

The simple but powerful model described here can be extended to analyze instabilities in other organs. It would also be interesting to couple the mechanics of this model with neural dynamics to better understand the neurodegenerative aspects of achalasia.

IV. ACKNOWLEDGMENTS

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Appendix: Methods

All data and measurements in this study were taken by Northwestern University. All human participant data were collected and handled in compliance with applicable ethical standards, institutional policies, and relevant regulations. Ethical approval was obtained where required, and informed consent was obtained from all participants or appropriately.

The manometer used had 36 pressure sensors and 18 impedance sensors. Between 7 and 12 swallows were identified per patient (278 swallows across the 29 type III achalasia patients), and every swallow was analyzed. For each swallow, a rectangular box was manually selected in the time–sensor plane around the region of peak pressure (Fig. 1B). The state of the swallow, (λ_1, λ_2) , was defined as the mean of the corresponding quantities over all sensors and all time points within the box, and the patient state was defined as the mean over that patient’s swallows. To quantify the variability within a swallow, the sensor-averaged value was first computed at each time

point in the box, and the standard deviation of these sensor-averaged values over the duration of the box was then taken; these are the error bars shown in Fig. 8A. A patient was classified as predicted to corkscrew if the patient-mean state lies on the unstable side of the marginal stability curve; the fraction of individual swallows past the curve (Fig. 8B) and the receiver operating characteristic obtained by thresholding the patient-mean signed distance to the curve (Fig. 8C) are reported as robustness checks. Classification metrics are reported at the patient level because swallows within a patient are not independent. Impedance signals were used to estimate luminal diameter by converting impedance to conductance and separating the portion attributable to the esophageal wall (“tissue conductance”) from the portion due to intrabolus fluid. Using established calibration constants for electrode spacing (20 mm) and bolus conductivity (0.812 mS/mm) [26], intrabolus conductance was mapped to cross-sectional area, after which the catheter area was added back. The luminal radius was then calculated as $r = \sqrt{\text{CSA}/\pi}$, providing a dynamic estimate of the radius of the esophageal opening independent of pressure. Because the 36 pressure sensors and 18 impedance electrodes are spaced differently along the catheter, the same segment of the esophageal body corresponds to pressure channels 10–30 and to impedance channels 5–15; for the spatially averaged time traces in Fig. 1D and H and for the resting inner radius in Fig. 6, the pressure and cross-sectional area were averaged over these respective channel ranges at each time point. The contraction parameter λ_1 was then found from the measured pressure and the calculated inner radius as described in the main text.

In TBE trials, a low-density barium sulfate suspension was orally ingested over the course of 15–20 s and radiographs were taken; the barium suspension acts as a radiocontrast agent and is thus visible in the images. Typically, only one set of TBE radiographs for a single swallow was taken per patient. Image segmentation was performed by hand-drawing a line along the midline of the barium column and using software to represent the curve. The axial stretch λ_z was defined as the arc length of this centerline during the swallow (a helix in corkscrew patients) divided by the length of the centerline at rest, when the esophagus is a straight line from its top to its bottom. Each radiograph included a ruler of known length provided by the clinicians, which was used to convert pixel distances to physical lengths, and images of the same patient at rest and during the swallow were registered with reference to a common vertebral landmark. The number of helical turns $\tilde{k}$ was counted directly on the swallow esophagram, and a patient was assigned to the corkscrew group if $\tilde{k} \neq 0$.

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